HITESH LENIN

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PROFESSIONAL SUMMARY

- **Results-driven AI/ML Engineer and Python Developer** with over **11 years of experience** in designing, developing, and deploying enterprise-grade **AI/ML, data engineering, and cloud-based applications** across manufacturing, finance, healthcare, and retail domains.
- AI/ML Architect with strong experience designing, building, and deploying production-grade machine learning and **Generative AI** solutions, including Retrieval-Augmented Generation (**RAG**) and **agentic AI** workflows.
- Built **multi-agent and single-agent systems** following patterns like planner–executor, tool-augmented reasoning, and self-validation to handle complex user requests reliably.
- Architected and deployed multi-agent AI systems using **LangGraph** for stateful orchestration, enabling autonomous task decomposition, parallel agent execution, and dynamic routing based on task complexity.
- Hands-on experience implementing agent-driven AI patterns for multi-step task orchestration, tool invocation, retrieval coordination, and response validation to improve accuracy and reduce hallucinations.
- Applied **prompt engineering, dynamic chaining, and agent-based architectures** using LangChain to enhance response accuracy and explainability.
- Implemented **Model Context Protocol (MCP)** integrations to securely connect LLM agents with internal tools and data sources (APIs, databases, document stores), enabling standardized, reusable tool access.
- Proficient in integrating LLMs (OpenAI, Hugging Face) into production using **FastAPI and Flask**, with hands-on experience in MLOps, CI/CD, and model monitoring.
- Experience with frontier models like GPT-4o, Claude 3.5, and **Llama 3.x/4** specifically for tool-calling and JSON-mode outputs.
- Experienced in designing and deploying **NLP pipelines** for sentiment analysis, document summarization, named entity recognition (NER), and translation using **Hugging Face Transformers, spaCy, and GPT-based architectures**.
- Hands-on experience in **computer vision**, building and deploying object detection, OCR, and image segmentation solutions using **OpenCV, TensorFlow, and Keras** for real-time visual inference.
- Highly skilled in **Python development** for data manipulation, transformation, and analytics using **Pandas, NumPy, Scikit-learn, PySpark, and Matplotlib**, with proven success in building **scalable ETL pipelines** on **Azure Databricks** and **AWS Glue** to process multi-terabyte datasets via batch and streaming ingestion.
- Experienced in developing and deploying **GenAI and machine learning solutions**, including **LLMs, Retrieval-Augmented Generation (RAG), LangChain orchestration**, and **Hugging Face Transformer fine-tuning**, to deliver predictive intelligence and automate business workflows.
- Proficient in implementing **CI/CD pipelines** using **Azure DevOps, Jenkins, Terraform, Docker, and Kubernetes**, and in establishing **data validation frameworks** with **PySpark** and **Great Expectations** to ensure data accuracy and integrity.
- Extensive experience leveraging **Azure (ADF, Synapse, Event Hubs, Purview, Delta Live Tables)** and **AWS (S3, EMR, Redshift, Kinesis, Lambda, CloudFormation, CloudWatch)** for secure, compliant, and cost-optimized data ecosystems.
- Skilled in using **AI-assisted development tools** such as **GitHub Copilot** for intelligent code generation, testing, and documentation, enhancing development efficiency and quality assurance.
- Recognized for strong collaboration with **cross-functional ML, DevOps, and product teams** in **Agile/Scrum environments**, leading **cloud migration, data governance, and compliance initiatives** (GDPR, PII masking) to build reliable, scalable, and insight-driven enterprise data solutions.

TECHNICAL SKILLS

Programming Languages	Python, SQL, JavaScript, Scala, Shell Scripting
Python Libraries & Frameworks	Pandas, NumPy, PySpark, Scikit-learn, Matplotlib, Great Expectations, LangChain, Hugging Face Transformers, TensorFlow, PyTorch
AI / ML & GenAI	Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, Prompt Engineering, Model Fine-Tuning, Generative AI Pipelines, SageMaker, Bedrock.
Data Engineering & ETL	Azure Data Factory (ADF), Databricks, AWS Glue, Apache Spark, Hive, Sqoop, Kafka, Structured Streaming, Delta Live Tables
Cloud Platforms	**Azure:** Data Lake Gen2, Synapse Analytics, Event Hubs, Purview, Delta Lake, Databricks **AWS:** S3, EMR, Redshift, Kinesis, Lambda, CloudFormation, CloudWatch
Databases & Storage	SQL Server, PostgreSQL, MySQL, MongoDB, HDFS, Redshift, Oracle 11g, NoSQL Databases
Data Modeling & Formats	Star & Snowflake Schemas, Dimensional Modeling, Parquet, Avro, ORC, RC, JSON, CSV
DevOps & CI/CD	Azure DevOps, Jenkins, GitHub, Terraform, Docker, Kubernetes, Databricks CLI
Web & Microservices	Flask, Django, Node.js, Express.js, REST APIs, React, Redux, Angular, HTML5, CSS3
Data Visualization & BI	Power BI, Tableau, Excel (Pivot Tables, DAX, Power Query), SSRS
Version Control & Collaboration	Git, GitHub, JIRA, Confluence, Agile / Scrum
Automation & Productivity Tools	GitHub Copilot, Celery, Redis, Bash / Shell Automation, PowerShell
Data Governance & Security	Azure Purview, Unity Catalog, IAM, PII Masking, GDPR Compliance, Role-Based Access Control (RBAC)

EDUCATION & CERTIFCATIONS

- Master of Science in Business Analytics and Information Systems.
- Bachelor's in Electronics and Communication Engineering.
- Certified Python Developer (PCAP)
- Microsoft Certified: Python for Data Science
- Azure AI Fundamentals
- AWS Certified AI Practitioner
- Databricks Certified Data Engineer

PROFESSIONAL EXPERIENCE

HUBBELL INCORPORATED, CLEVELAND, OH

Domain: Manufacturing

AI/ML Engineer (Agentic AI, ML Focused)

Aug 2024 – Present

- Translated data science prototypes into **production-grade ML services** using **Python, TensorFlow, PyTorch, and XGBoost**, deploying end-to-end pipelines with reproducibility, versioning, and automated testing baked into every release.
- Built scalable **model serving infrastructure** using **FastAPI** with 25+ inference endpoints, implementing batching strategies, latency optimization, and Pydantic schema validation serving 100,000+ daily requests with consistent SLA adherence.
- Implemented full **model lifecycle tooling** using **AWS SageMaker Model Registry** and MLflow for experiment tracking, model versioning, and deployment automation — enabling one-click promotion from staging to production with full audit trails.
- Designed **online and offline inference pipelines** leveraging **AWS SageMaker Processing and Training** jobs with automated retraining triggers, blue-green deployment strategies, and containerized model packaging via **Docker and Kubernetes**.

- Owned **production health** of ML systems including drift detection, performance regression monitoring, and rollback strategies using **CloudWatch** dashboards and custom alerting, ensuring rapid incident response and minimal model degradation.
- Engineered **metadata-driven ETL frameworks** in **PySpark** and Python collaborating with Data Engineering on feature pipelines and data contracts, enabling reuse across 100+ data sources with schema validation via **Great Expectations**.
- Applied **MLOps practices** end-to-end using **Azure DevOps, Jenkins, Docker, and Kubernetes** — automating build, test, and deployment of ML services with CI/CD pipelines enforcing code quality, dependency management, and vulnerability checks.
- Architected **vector database infrastructure** using **Pinecone and Milvus** with HNSW indexing and quantization techniques for billion-scale semantic search, optimizing throughput and memory footprint for production RAG workloads.
- Implemented **GPU acceleration** for model training with hyperparameter tuning, feature engineering, and distributed training patterns, significantly reducing training time and improving model accuracy across classification and clustering tasks.
- Ensured **responsible AI and compliance** standards through PII masking, GDPR-aligned data governance using **Azure Purview and Unity Catalog**, RBAC controls, and audit logging across all ML pipeline components.

Environment: *Python, PySpark, Pandas, NumPy, Delta Lake, Azure Databricks, NLP, LLMs, RAG, Delta Live Tables, Unity Catalog, XGBoost, Power BI, SQL, GraphQL, Hugging Face Transformers, PyTorch, LangChain, GPT-4, GitHub, AWS Lambda, ECS, EKS, FAISS, Docker, Kubernetes, Terraform, Jenkins, Azure DevOps, Great Expectations, Boto3, AWS CloudFormation, AWS Sagemaker, Keras.*

Allstate, Chicago, IL

Domain: Insurance

Python Full-stack Developer (ML Focused)

Sept 2022-Jul 2024

- Architected and developed a full-stack insurance analytics platform using **Python (FastAPI)** and **Vue.js**, enabling real-time policy, claims, and underwriting insights with a microservices-driven architecture.
- Built high-performance RESTful and asynchronous APIs using FastAPI with **OAuth2 authentication and OpenAPI standards**, supporting claims processing, risk scoring, and customer policy management workflows.
- Designed and implemented event-driven data ingestion pipelines using **Apache Kafka and Apache Spark (PySpark)** to process real-time telematics and claims data for risk analysis and fraud detection.
- Developed predictive models for **claim severity and claim frequency** using Gradient Boosting and CatBoost, improving underwriting decision accuracy and reducing loss ratio exposure.
- Implemented a fraud detection engine using **graph-based anomaly detection (NetworkX)** and ensemble ML techniques, identifying suspicious claim patterns such as duplicate claims and staged accidents.
- Built NLP-based document processing pipelines using **spaCy and BERT embeddings**, automating classification and extraction of information from insurance documents such as FNOL reports, adjuster notes, and policy contracts.
- Deployed machine learning pipelines using **Google Cloud Vertex AI**, including automated training, model versioning, and scalable batch and real-time prediction services.
- Designed feature stores using **Feast**, enabling consistent feature access across training and inference pipelines while reducing training-serving skew in production models.
- Implemented real-time model inference using **gRPC-based services**, optimizing latency for fraud scoring and underwriting risk predictions in high-throughput environments.
- Built end-to-end data pipelines using **dbt and BigQuery**, transforming raw insurance data into curated datasets for analytics, reporting, and ML model consumption.
- Established model monitoring and drift detection using **Evidently AI**, tracking statistical data changes and performance degradation across claims and fraud models with automated retraining triggers.
- Integrated CI/CD pipelines using **GitLab CI/CD** and container orchestration with **Docker Swarm**, enabling automated build, test, and deployment cycles for both application and ML services.
- Implemented secure data governance and compliance controls aligned with **HIPAA and SOC2 standards**, including encryption, access control policies, and audit logging for sensitive customer and claims data.
- Optimized application performance using **Redis caching, asynchronous processing, and query optimization in**

BigQuery, improving response times for high-volume insurance transactions and analytics queries.

Environment: Python, FastAPI, Vue.js, JavaScript, HTML5, CSS3, PySpark, Apache Kafka, BigQuery, dbt, Feast, NetworkX, spaCy, BERT, Vertex AI, Redis, Docker Swarm, GitLab CI/CD, Evidently AI, OAuth2, gRPC, Linux

THOMSON REUTERS, EAGAN, MN

Domain: Finance

Python Full-stack Developer (ML Focused)

Aug 2021-Aug 2022

- Designed front end and backend of the application using Python on Django Web Framework and AngularJS developed consumer-based features and applications using **Python and Django** in test driven Development and pair-based programming.
- Developed Python backend services exposing **REST API** endpoints consumed by **React/Node.js** frontends, enabling seamless human-in-the-loop interaction with AI agent workflows.
- Implemented time-series forecasting models using **ARIMA** and **Prophet** for quarterly revenue projection and cash flow optimization, improving forecast accuracy by 18% compared to traditional statistical methods.
- Built document classification system using scikit-learn **TF-IDF** and **Naive Bayes** categorizing 100,000+ financial documents (loan agreements, tax forms, statements) into 25 document types.
- Developed web applications and RESTful web services and APIs using **Python Flask, Django, Pyramid, and PHP**.
- Developed RESTful APIs using Flask with JWT authentication and role-based access control (RBAC) exposing 40+ endpoints for portfolio analysis, transaction processing, and customer data retrieval.
- Shipped **ML models to production on AWS SageMaker**, building training pipelines with automated hyperparameter tuning (Bayesian optimization, 50+ combinations), model registry integration, and inference endpoints with measurable latency and throughput SLAs.
- Implemented **scalable model serving** with multi-variant endpoints for A/B testing (XGBoost vs LightGBM) on 10% production traffic, establishing data-driven model selection workflows that reduced false positives by 12%.
- Built **reproducible ML pipelines** with full versioning, automated PyTest-driven testing, and Docker-based packaging — ensuring training-serving consistency and eliminating data leakage risks through strict feature pipeline isolation.
- Developed **fraud detection system** using Isolation Forest and autoencoders (**TensorFlow/Keras**) with real-time inference via REST API, implementing drift monitoring on transaction pattern distributions to detect model degradation in production.
- Integrated **model monitoring and observability** tracking precision, recall, and AUC over time across loan default prediction and churn models, with automated alerts triggering retraining workflows when performance regression thresholds were breached.
- Collaborated with Data Engineering to design **feature pipelines** for customer churn prediction using Random Forest — engineering features from transaction history, demographics, and service interactions with defined data contracts to prevent training-serving skew.
- Applied **CI/CD and DevOps practices to ML workloads** using Jenkins and Kubernetes, automating model retraining, registry promotion, and deployment with rollback strategies and incident response runbooks for production ML services.
- Enforced **responsible AI and compliance requirements** implementing PCI DSS-compliant data handling, PII masking in feature stores, JWT-based RBAC across all ML service endpoints, and dependency vulnerability scanning in every CI pipeline run.

Environment: *Python, Django, Flask, AngularJS, HTML5, CSS3, JavaScript, Pandas, NumPy, Scikit-learn, AWS (SageMaker, EMR, S3, Redshift), Azure Databricks, PostgreSQL, MongoDB, SQLAlchemy, Kubernetes, Jenkins, Git, Jira, PyTest, Unix/Linux*

CYBAGE SOFTWARE PVT LIMITED, HYDERABAD, INDIA

Domain: Healthcare

Data Engineer

Jan 2016 – Sept 2019

- Designed and maintained large-scale ETL pipelines on **AWS** using **S3, EMR (Hadoop/Spark), and Redshift** to process healthcare claims, **EHR/EMR** data, eligibility files, encounter data, lab results, and clinical documentation.
- Built **PySpark** and Scala Spark jobs on **AWS EMR** to transform high-volume healthcare datasets for claims adjudication, quality-of-care reporting, HEDIS measure calculations, and risk-adjustment workflows.
- Developed optimized Hive tables (ORC/Parquet, partitioned, bucketed) on **EMR/Hive** Metastore to support cost/utilization analysis, provider performance analytics, and population health insights.
- Ingested and normalized complex healthcare data formats including **HL7**, X12 837/835 claims, CCD/CDA documents, and **FHIR** bundles using Spark, custom parsers, and **AWS** storage layers.
- Designed and optimized **scalable ML data pipelines** capable of processing high-volume, high-dimensional datasets using Python, Pandas, NumPy, Spark, and cloud-native workflows.
- Integrated **MongoDB and HBase** datasets into EMR pipelines to process unstructured clinical notes, physician narratives, radiology reports, and discharge summaries for downstream analytics.
- Developed **PL/SQL** and SQL scripts to generate regulatory reports, patient risk profiles, chronic condition registries, ICD/SNOMED mappings, and utilization trend dashboards.
- Enhanced **Redshift** schemas and fact/dimension models to support healthcare star-schema reporting, enabling fast insights for claims trends, care management, readmission analysis, and cost forecasting.
- Supported production **ETL** workflows using **Ab Initio** and EMR-based Spark jobs, troubleshooting failures, optimizing cluster usage, and ensuring SLA adherence for mission-critical healthcare reporting cycles.
- Collaborated with data architects and clinical SMEs to implement **HIPAA-**compliant data pipelines, including **PHI** masking, secure S3 bucket policies, IAM controls, and audit logging across AWS components.

Environment: *AWS (S3, EMR, Redshift, IAM), Hadoop, Spark (PySpark, Scala), Hive, ORC, Parquet, MongoDB, HBase, Python, Pandas, NumPy, SQL, PL/SQL, Ab Initio, HL7, X12 (837/835), CCD/CDA, FHIR, ICD, SNOMED, Unix/Linux*

INDIUM SOFTWARE, CHENNAI, INDIA

Domain: Retail

Data Analyst

Jun 2013 – Dec 2015

- Designed and developed **50+ Tableau and Power BI dashboards** to track KPIs across sales, marketing, and operations, improving executive decision-making and performance visibility.
- Performed **data mining and analysis using SQL, Excel, and Python (Pandas, NumPy)**, driving actionable insights for customer segmentation and targeted marketing campaigns.
- Automated **data extraction and cleansing processes** using **Python scripts and shell automation**, reducing manual reporting time by **over 60%**.
- Created and optimized **dimensional data models** (Star and Snowflake schemas) to support scalable reporting and analytics.
- Built **Excel VBA macros and advanced formulas** to automate recurring reporting tasks and data reconciliations across departments.
- Developed **interactive Tableau dashboards** with parameters, trend analysis, and forecasting models to identify seasonal sales patterns and improve revenue predictions.
- Integrated **data from multiple SQL sources, CRM systems, and Google Analytics** for consolidated customer behavior and campaign performance analysis.
- Defined and implemented **custom KPIs and DAX measures in Power BI**, enabling teams to monitor conversion rates, churn, and product performance in real time.
- Collaborated with business users to document reporting requirements, validate data accuracy, and enhance visualization usability through intuitive designs and tooltips.
- Conducted **A/B testing and statistical analysis in R and Python**, providing insights into the effectiveness of marketing and promotional strategies.

Environment: *SQL Server, Tableau, Power BI, Excel (VBA, Macros), Python 3.x, R, Shell Scripting, Google Analytics, CRM Systems, Star & Snowflake Schema Modeling, SSRS, Power Query, DAX*